Adding power series

It's important for later lessons on series solutions for differential equations that we be able to add power series.

Conditions for addition

Power series can only be added when

1. their indices start at the same values, and
2. the powers of x in each series are "**in phase**," which means that both series start with the same power of x.

Remember that the **index** of a power series is given by the small equation we see below the summation notation, in this case, $n = 0$:

$$\sum_{n=0}^{\infty} \frac{5^n}{n!} x^n$$

We'll usually use n or k for the indexing variable, and the index most commonly (but certainly not always) begins at 0 or 1.

So when we say that the indices of separate power series need to match before we can add them, we're saying that they both need to start at the same value. These series have matching indices $n = 0$,

$$\sum_{n=0}^{\infty} \frac{5^n}{n!} x^n \qquad \sum_{n=0}^{\infty} \frac{(-1)^n}{9^n} x^{2n+1}$$

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while these series have indices that don't match, since one is $n = 0$ while the other is $n = 1$.

$$\sum_{n=0}^{\infty} \frac{5^n}{n!} x^n \qquad \sum_{n=1}^{\infty} \frac{(-1)^n}{n} x^n$$

We also need to make sure that the powers of x in each series are in phase. For example, we can't add a series whose first term is for x^1 to a series whose first term is x^2. Two series will only be in phase if they both start with x^0, both start with x^1, both start with x^2, etc.

How to add series

Let's assume that two series aren't in phase and that their indices start at different values. In that case, we'll start by rewriting the series so that they're in phase, and then we'll rewrite that result to make the indices match.

Once both conditions are met, we can add the series term by term, simplifying the end result as much as possible.

Let's do an example so that we can see these steps in action.

Example

Find the sum of the series.

$$\sum_{n=1}^{\infty} 2nc_n x^{n-1} + \sum_{n=0}^{\infty} 6c_n x^{n+1}$$

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Because the first series starts at $n = 1$, the first power of x for that series will be $x^{n-1} = x^{1-1} = x^0$. And because the second series starts at $n = 0$, the first power of x for that series will be $x^{n+1} = x^{0+1} = x^1$.

So the series are not in phase. To fix this, we'll pull the first term from the first series outside of the summation. That will force the index to start at $n = 2$ instead of $n = 1$.

$$2(1)c_1x^{1-1} + \sum_{n=2}^{\infty} 2nc_nx^{n-1} + \sum_{n=0}^{\infty} 6c_nx^{n+1}$$

$$2c_1 + \sum_{n=2}^{\infty} 2nc_nx^{n-1} + \sum_{n=0}^{\infty} 6c_nx^{n+1}$$

Written this way, both series begin with an x^1 term, so the series are now in phase. But the indices are $n = 2$ and $n = 0$ and don't match. To fix this, we'll look at the exponents of x, setting $k = n - 1$ in the first series and $k = n + 1$ in the second series.

Plugging the index of the first series $n = 2$ into $k = n - 1$ gives $k = 1$, and plugging the index of the second series $n = 0$ into $k = n + 1$ gives $k = 1$. And solving both $k = n - 1$ and $k = n + 1$ for n gives $n = k + 1$ and $n = k - 1$, respectively.

Making all these substitutions, the sum becomes

$$2c_1 + \sum_{k=1}^{\infty} 2(k+1)c_{k+1}x^k + \sum_{k=1}^{\infty} 6c_{k-1}x^k$$

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Now that the indices match and the series are in phase, we can add them.

$$2c_1 + \sum_{k=1}^{\infty} 2(k+1)c_{k+1}x^k + 6c_{k-1}x^k$$

$$2c_1 + \sum_{k=1}^{\infty} (2(k+1)c_{k+1} + 6c_{k-1})x^k$$

We can leave this in k, but since the original sum was given in terms of n instead of k, let's put this back in terms of n.

$$2c_1 + \sum_{n=1}^{\infty} (2(n+1)c_{n+1} + 6c_{n-1})x^n$$

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